Nicola Muça Cirone

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Education

PhD Mathematics - CDT Mathematics of Random Systems, 2022-2025
Imperial College London - University of Oxford

- Deep Learning, Mathematical Finance, Rough Paths Analysis.
- Supervised by *Dr. Cristopher Salvi*.
- Researching scaling limits of Neural Networks
- Working in theoretical foundations for State Space Models.
- Developing new Kernel Methods for Time Series analysis.

MSc Mathematics, University of Pisa - ETH Zürich 2021-2022

- Honour's degree (110/110 cum Laude) | GPA 30 cum Laude / 30
- Stochastic Calculus, Stochastic PDEs, Mathematical Finance, ML in Finance.
- Thesis: *Kernel Methods on Path Spaces* supervised by *Prof. Josef Teichmann*.

BSc Mathematics, University of Pisa 2018-2021

- Honour's degree (110/110 cum Laude) | GPA 29.88/30
- Thesis on advanced topics in Geometrical Model Theory.

Experience

Intern - Quant Researcher, Quadrature Capital | London July - Sept 2025
Deep learning for financial markets.

Intern - ML Researcher, Cartesia AI | San Francisco Feb - May 2025
Research on new ways to distill LLMs to different architectures, depths and sizes.
Research on development of ViT models capable of multi-modal reasoning.

Industry Project - Neural Signature Kernels, Imperial College London June 2024
Co-organized 2024 CDT Industry Project and supervised PhD students
in applying Neural Signature Kernels to time series analysis.

Teaching Assistant - Stochastic Calculus for Finance, ICL Business School Oct - Dec 2023
MSc Risk Management & Financial Engineering. Taught exercise classes.

Industry Project - Brain signal analysis with Signatures, Imperial College London June 2023
Challenge: analyze MEG data from several subjects, detect the one who had a seizure.
Our team used Signature techniques, the success and clarity of our results won the competition.

Teaching Assistant - Numerical Methods for Finance, Imperial College London Feb 2023
MSc Mathematics and Finance. Taught exercise classes.

Reviewed Publications

[4] Benjamin Walker, Lingyi Yang, **Nicola Muca Cirone**, Cristopher Salvi and Terry Lyons.
Structured Linear CDEs: Maximally Expressive and Parallel-in-Time Sequence Models.
NeurIPS 2025 (Spotlight)
url: https://arxiv.org/abs/2505.17761

[3] Sajad Movahedi, Felix Sarnthein, **Nicola Muca Cirone** and Antonio Orvieto.
Fixed-Point RNNs: From Diagonal to Dense in a Few Iterations
NeurIPS 2025 (Spotlight)
url: https://arxiv.org/abs/2503.10799

[2] **Nicola Muca Cirone**, Antonio Orvieto, Benjamin Walker, Cristopher Salvi and Terry Lyons.
Theoretical Foundations of Deep Selective State-Space Models.
NeurIPS 2024
url: https://arxiv.org/abs/2402.19047

[1] **Nicola Muca Cirone**, Maud Lemercier and Cristopher Salvi.
Neural signature kernels as infinite-width-depth-limits of controlled ResNets.
Proceedings of the 40th International Conference on Machine Learning, 25358–25425, 2023.
url: https://proceedings.mlr.press/v202/muca-cirone23a.html

PREPRINTS

[5] **Nicola Muca Cirone**, Owen Futter and Blanka Horvath.
Kernel Learning for Mean-Variance Trading Strategies
url: https://arxiv.org/abs/2507.10701

[4] **Nicola Muca Cirone** and Cristopher Salvi.
ParallelFlow: Parallelizing Linear Transformers via Flow Discretization.
url: https://arxiv.org/abs/2504.00492

[3] Mie Glückstad, **Nicola Muca Cirone** and Josef Teichmann.
Signature Reconstruction from Randomized Signatures.
Submitted to Journal
url: https://arxiv.org/abs/2502.03163

[2] **Nicola Muca Cirone** and Cristopher Salvi.
Rough kernel hedging.
url: https://arxiv.org/abs/2501.09683

[1] **Nicola Muca Cirone**, Jad Hamdan and Cristopher Salvi.
Graph Expansions of Deep Neural Networks and their Universal Scaling Limits.
Submitted to Journal
url: https://arxiv.org/abs/2407.08459

TALKS

Oxford-ETH Workshop in Mathematical Finance , Oxford University — June 2025
Title: *Signature Reconstruction from Randomized Signatures*.

Workshop on Recent Developments in Theoretical Machine Learning , Imperial-X — Jan 2025
Title: *Graph Expansions of Deep Neural Networks*.

Stochastic Analysis Seminar , ShanghaiTech University — Nov 2024
Title: *Theoretical Foundations of Deep Selective State-Space Models*.

DataSig - Rough Paths Interest Group , Alan Turing Institute — May 2024
Title: *Signature Reconstruction from Randomized Signatures*.

Computer Vision Group Seminar , University of Michigan — April 2024
Title: *Theoretical Foundations of Deep Selective State-Space Models*.

Talks in Financial and Insurance Mathematics , ETH Zürich — Dec 2023
Title: *Neural Signature Kernels (and Trees)*.

7th London-Paris Bachelier Workshop, Imperial College London — Sept 2023
Title: *Rough Kernel Hedging*.

AWARDS

NeurIPS 2024 Scholar Award, Neural Information Processing Systems Foundation — 2024

INDAM Scholarship, Istituto Nazionale di Alta Matematica — 2022
Two year scholarship from the Italian Institute of Higher Mathematics.
Placed 2nd at National level.

COMPETENCES

Languages Italian (*native*), English (C1), French (*basic*)

Technical Skills Python, Jax, Pytorch, Triton, CUDA, R, Matlab, Java